

SDG Goal 8 Decent work and economic growth

SDG Target 8.4 Improve progressively, through 2030, global resource efficiency in consumption and production and endeavour to decouple economic growth from environmental degradation, in accordance with the 10-Year Framework of Programmes on Sustainable Consumption and Production, with developed countries taking the lead

SDG Indicator 8.4.1 Material footprint, material footprint per capita, and material footprint per GDP

Time series Raw material consumption (RMC)

1. General information on the time series

- Date of national metadata: July 10, 2022
- National data: <https://sdg-indicators.de/8-4-1>
- Definition: The time series measures the amount of raw material used globally for the production of goods for domestic consumption, gross fixed capital formation, changes in inventories and acquisitions less disposals of valuables. The time series is also available in relation to the domestic population and real gross domestic product (GDP).
- Disaggregation: Not available.

2. Comparability with the UN metadata

- Date of UN metadata: March 2025
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-08-04-01.pdf>
- The time series is not compliant with the UN metadata, but provides additional information. The RMC calculation is based on a hybrid model using domestic input-output tables, life cycle information and other data sources. In the UN metadata, a multi-regional input-output (MRIO) framework is mentioned. Further methodological differences may exist e.g. due to deviations in the treatment of secondary raw materials.

3. Data description

- The data on raw material consumption is generated based on a hybrid estimation model combining physical and monetary information from official and other data sources. These calculations are project work run by the Environmental-Economic Accounts unit of the Federal Statistical Office.

The data on GDP is calculated by the Federal Statistical Office's National Accounts as a secondary statistic. GDP is adjusted based on a price base changing every year (previous year's price base). After several revisions due to new data input, final results are available four years after the first preliminary release.

4. Access to data source

- Statistical Report: Raw material equivalents (RMC) (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/rohstoffe-materialfluesse-wasser/_inhalt.html#sprg380298
- National accounts – Gross value added, gross domestic product (nominal/price-adjusted): GENESIS online 81000-0001:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=81000-0001&bypass=true&language=en>

5. Metadata on source data

- Raw materials for Germany: Analysis of consumption, investment and export at the macro and meso levels (only available in German):
<https://www.umweltbundesamt.de/publikationen/rohstoffe-fuer-deutschland>
- Quality report: Economic Accounts (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Umwelt/ugr.pdf?>
- Project report: Improving Raw Material Equivalent Estimation in Environmental-Economic Accounts:
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/rohstoffe-materialfluesse-wasser/Publikationen/Downloads/improving-raw-material-equivalents-2025-1_2025028.pdf

6. Timeliness and frequency

- Timeliness: Irregular
- Frequency: Annual

7. Calculation method

- Unit of measurement: Million tonnes, Tonnes per capita
- Calculation: Not applicable.